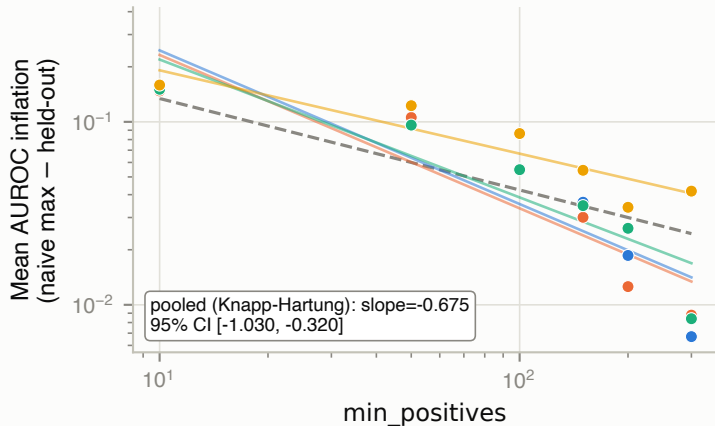


Figure 2. The vanishing sampling-noise bias shrinks as $n^{-1/2}$, confirmed across four scales and two independent routes

● ESM2-8M ● ESM2-35M ● ESM2-150M ● ESM2-650M - - - theoretical $n^{-1/2}$

A. Route 1 — held-out AUROC gap (real labels)



B. Route 2 — held-out AUROC gap (permuted labels)

